Network Topologies — Guided Notes ANSWER KEY

Unit 1.2.1 | CompTIA Network+ N10-009 Obj. 1.2 | N10-008 1.2

For instructor use / distribute after students complete the notes.

Question 1

In a _______ topology, all devices connect to a single shared cable called a _______. If that cable fails at any point, _______ devices lose connectivity.

Answer: Bus; backbone; all.

Question 2

A ring topology passes a special _______ frame around the loop to control who can transmit. The main weakness of a single-ring design is that one _______ or _______ failure breaks the entire loop.

Answer: Token; device; cable segment.

Question 3

In a star topology, all devices connect to a central _______. A single device failure affects only _______ device. The central device, however, is a _______ — if it fails, the entire segment goes down.

Answer: Switch (or hub); that one; single point of failure.

Question 4

In a full mesh topology with 10 devices, how many point-to-point links are required? Show the formula and the result.

Answer: Formula: n(n-1)/2. With n=10: 10 × 9 / 2 = 45 links.

Question 5

Explain the difference between a full mesh and a partial mesh. Why is partial mesh the more common real-world design?

Answer: Full mesh connects every device to every other device — maximum redundancy but exponential cost. Partial mesh only gives multiple connections to critical devices. Real-world networks use partial mesh because full mesh becomes prohibitively expensive at scale.

Question 6

_______ topology combines two or more topology types. Give one real-world example: _______.

Answer: Hybrid; e.g., star topology inside each building connected by a partial mesh WAN between buildings (or similar valid example).

Question 7

Physical topology is the _______ layout of cables and devices. Logical topology is the path _______ actually takes. These two can differ — give an example of when they do.

Answer: Actual physical; data. Example: Token Ring physically wired as a star through a MAU, but data travels logically in a ring. Or: two devices on the same physical switch but on different VLANs are *logically on separate networks.*

Question 8 — Real World Application

REAL WORLD: An entire floor of a building loses network access simultaneously. No individual device issues are reported. Based on what you know about network topologies, what topology is most likely in use, what single component probably failed, and what design change would prevent this from happening again?

Answer: Star topology is almost certainly in use — one switch failure taking down all devices is the classic star failure mode. The access-layer switch lost power or hardware failed. Prevention: add a redundant switch in a stacked or dual-uplink configuration, use a managed switch with dual power supplies, or implement a two-tier design where devices connect to a redundant distribution layer.